

VIT BHOPAL UNIVERSITY - PORTAL SUBMISSION

Course Component: Python Programming Lab

Project Title: FitTrack - Interactive Fitness Assistant

File Name: fittrack.py

Welcome to FitTrack!

FitTrack is a lightweight, interactive command-line fitness and health routine assistant written in Python. It simulates a professional health orientation experience right inside your terminal, complete with user profiling, dynamic fitness evaluations, tailored exercise suggestions, and automated summary report generation. The program is structured entirely around standard libraries, ensuring maximal compatibility and pure console performance without overhead.

Features

- **Intelligent Profile Builder:** Captures and processes user demographics including profile names, exact age metrics, and vital structural metrics.
- **Guarded Range Input Validation:** Features dynamic exception handling for user floating points to filter out corrupted text or unrealistic physiological metrics (e.g., negative weights).
- **AutomHealth Evaluation:** Calculates the user's Body Mass Index (BMI) dynamically and maps the result against standard medical range categories for adults.
- **Targeted Goal-Oriented Routing:** Offers 4 core target categories split into precise equipment and structural configurations:
 - **Gain:** Focuses on weight gain and strength building via heavy tools like dumbbells and barbells.
 - **Lose:** Focuses on calorie burning and weight loss via high-impact mat and floor routines.
 - **Maintain:** Focuses on general physical fitness using light resistance tools.
 - **Abs:** Focuses on core-centric abdominal isolation exercises using pull-up bars or mats.
- **Medical Precaution Flagging System:** Safely detects, stores, and appends critical safety guidelines for profiles containing pre-existing clinical records or mobility impairments.
- **Itemized Session Summary Receipt:** Generates a polished text-based terminal receipt timestamped via system clocks, detailing localized dates, core metrics, and final health assessments.

Getting Started

Prerequisites

Make sure you have Python 3.x installed on your system. The software relies exclusively on built-in libraries (calendar, datetime, and random), meaning no external dependencies or pip installations are required.

Running the App

1. Save the source script exactly as fittrack.py.

2. Open your operating system terminal, command prompt, or integrated development environment (IDE) console.
3. Launch the application by executing the command:

```
python fittrack.py
```

How to Use

1. **Authentication & Demographics:** Enter a profile nickname and input your age as a whole integer.
2. **Metric Configuration:** Provide your exact weight in kilograms and height in centimeters using standard decimal notation when prompted.
3. **Core Objective Selection:** Choose your fitness target by inputting a distinct numerical index from 1 to 4.
4. **Medical Screening:** Type 'yes' or 'no' to report health issues. If yes, write a brief textual description for screening context.
5. **Evaluate Output:** Read the dynamically generated workout plans, safety precautions, and biological category analysis.
6. **Multi-Session Loop:** Enter 'yes' when asked if you want another session to reset and log a new profile, or enter any other character to exit.

Code Structure Overview

- **read_number(prompt, minimum, maximum):** Implements a validation while loop that handles string parsing (.isnumeric()), traps text errors, and strictly binds numerical entries between specified constraints.
- **calculate_bmi(weight, height):** Performs physical unit conversion (centimeters to meters) and executes standard mass/height-squared calculations to yield precise float ratios.
- **get_suggestions(bmi, age, goal, has_medical_history, medical_history):** Houses conditional trees (if-elif-else) that check age boundaries, evaluate BMI ranges, assign exercise names, and inject medical safety overrides.
- **profile():** Acts as the primary data-collection sub-routine, organizing interactive prompts and returning a compiled dictionary entity containing the current user state.
- **Global Main Loop:** Orchestrates the session loop execution flow, instantiates localized datetime objects, applies random welcome phrases, and prints the final system receipt.